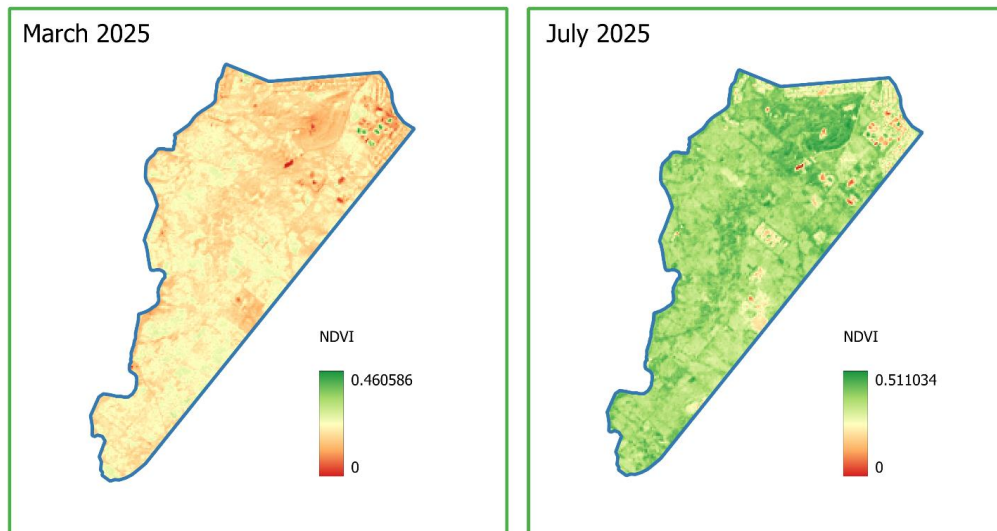


Final Project

Seasonal NDVI Assessment of the Grunewald (“Green Forest”)



(A very rough) map showing NDVI comparison in the Grunewald between March and May 2025

Objective

Produce a quantitative and cartographic comparison of vegetation health in the Grunewald forest between two Landsat (8 and 9) scenes taken in different seasons. This project combines skills from (almost) every lesson: digitizing (Lesson 4), coordinate reference systems (Lesson 2), virtual rasters and band math (Lesson 5), and print layout design (Lesson 1).

Data

- Landsat scene, March 2025 (newly provided for this project)
- Landsat scene, July 2025 (already provided in Lesson 5)
- [Link to the dataset](#) (download this PDF file and then click)
- Both scenes contain individual GeoTIFF bands, including Band 2 (Blue), Band 3 (Green), Band 4 (Red), and Band 5 (Near-Infrared).

Suggested Workflow

1. Digitize a polygon AOI (“area of interest”) covering the Grunewald. Create the new shapefile layer directly in, or reproject it to, EPSG:32633 (UTM Zone 33N — the correct UTM zone for Berlin).
2. Open the AOI's attribute table and use the Field Calculator with the **\$area** expression to add an area field. Record the resulting area.
3. For each scene (March and July), use Build virtual raster to stack bands 4, 3, 2 in that order (checking "Place each input file into a separate band") to create a True Colour Composite (TCC).
4. Compute NDVI — $(B5 - B4) / (B5 + B4)$ — on the full extent of each scene separately, using the Raster Calculator. Do this once for the March scene and once for the July scene.
5. Clip each TCC virtual raster and each full-scene NDVI raster to the Grunewald AOI using Clip Raster by Mask Layer (four clipped outputs in total).

6. Symbolize: the clipped TCC rasters as Multiband color with a Min/Max contrast stretch; the clipped NDVI rasters as Singleband pseudocolor with an appropriate colorbar.
7. Run Processing Toolbox → Raster Analysis → Raster Layer Statistics on each clipped NDVI raster. Save/screenshot the HTML report produced for each.
8. Build a single Print Layout containing all four clipped rasters (TCC July, TCC March, NDVI March, NDVI July) over a cool basemap (Hint: ESRI Standard), with title, legend, scale bar, north arrow, source, and the AOI outline — the final map-making exercise.

Questions (answer briefly, in your submission)

1. What is the AOI area (km²), computed via \$area in EPSG:32633? Why is UTM Zone 33N the correct choice for calculating areas in this part of the world, in general?
2. Report the mean, minimum, and maximum NDVI for the March scene and the July scene, taken from the Raster Layer Statistics HTML output. Include a screenshot of each report.
3. By how much (absolute difference) does mean NDVI change between March and July?
4. What does this suggest about coniferous versus deciduous forest cover in the Grunewald?

Submission Deliverables

- Deadline: **July 31, 2026**
- One PDF report containing:
 - Your name, matriculation number, and email address
 - all four maps — TCC and NDVI for both March and July — with title, legend, scale bar, north arrow, source, and the AOI outline.
 - Screenshots of both Raster Layer Statistics HTML reports.
 - A short written response (half a page) to the four questions above.

Grading Rubric

Criterion	Weight	Description
AOI & CRS handling	20%	AOI correctly digitized; reprojected to/created in EPSG:32633; area computed via \$area and reported correctly.
TCC virtual raster	15%	Virtual raster correctly built for each date using bands 4-3-2 in order, with separate-band stacking enabled; correctly stretched/symbolized.
NDVI workflow	25%	NDVI computed on full scene extent for both dates before clipping; correct clip-by-mask; correct pseudocolor symbology.
Statistics & interpretation	15%	Raster Layer Statistics run correctly on both clipped NDVI rasters (HTML screenshots included); seasonal change correctly quantified and interpreted.
Cartographic output	25%	Single layout with TCC and NDVI maps for both dates; title, legend, scale bar, north arrow, source, and AOI outline all present and correct.